Transient Heat Transfer Model with Erosive Ablation for Rocket Engines

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Nomenclature

Latin Symbols

A	Area [m²]
c_p	Specific heat at constant pressure [J/(kg·K)]
C_{adv}	Advective numerical coefficient [s⁻¹]
C_{diff}	Diffusive numerical coefficient [s⁻¹]
D	Diameter [m]
e	Layer thickness [m]
f_β	Aerodynamic shape factor for velocity mismatch [-]
G	Thermal conductance [W/(m²·K)]
h	Convective heat transfer coefficient [W/(m²·K)]
H	Enthalpy [J/kg]
k	Thermal conductivity [W/(m·K)]
L	Length [m]
M	Mach number [-]
$\dot{m}$	Mass flow rate [kg/s]
N	Number of injection orifices [-]
Nu	Nusselt number [-]
P	Pressure [Pa]
Pr	Prandtl number [-]
q	Heat flux [W/m²]
Q^*	Effective heat of ablation [J/kg]
r	Radial coordinate [m] / Recovery factor [-]
R_g	Specific gas constant [J/(kg·K)]
Re	Reynolds number [-]
s	Inner moving boundary position [m]
S	Equivalent slot thickness [m]

$\dot{s}$	Recession velocity [m/s]
t	Time [s]
T	Temperature [K]
u	Liquid injection velocity [m/s]
v	Free-stream gas velocity [m/s]

Greek Symbols

α	Thermal diffusivity [m²/s]
β	Velocity ratio (free-stream to injection) [-] / Numerical safety factor [-]
γ	Specific heat ratio [-]
Δ	Spatial or temporal discrete step
ϵ	Emissivity [-]
μ	Dynamic viscosity [Pa·s]
ξ	Transformed Landau spatial coordinate [-]
ρ	Density [kg/m³]
σ	Stefan-Boltzmann constant [5.670374×10^{-8} W/(m²·K⁴)]
ϕ	Stechman turbulent mixing parameter [-]
ω	Viscosity power-law exponent [-]

Subscripts

1	Inner ablative liner
2	Outer metallic casing
abl	Ablation threshold
am	Arithmetic mean (Eckert reference state)
aw	Adiabatic wall
ch	Chamber (stagnation conditions)
$cond$	Conduction
eff	Effective (film-cooled state)
ext	External ambient

$film$	Coolant film layer
$init$	Initial injection state
pc	Pseudo-critical thermodynamic state
r	Recovery (turbulent boundary layer)
rad	Radiation
t	Throat
∞	Free-stream / Static conditions

1. Introduction

One of the most critical aspects of a rocket engine is its ability to withstand extreme aerothermodynamic loads. In solid rocket motors or liquid rocket engines utilizing ablative or erosive cooling, a liner is strictly required to prevent the outer metal casing from exceeding its yield strength under maximum operating pressure.

In practical applications, the liner functions as an ablative barrier. Once the inner surface temperature reaches the material's ablation threshold, the liner begins to erode, gradually thinning at a rate driven by the net heat flux. This phase change and mass removal process dissipates heat, preventing structural failure.

This high-fidelity model advances beyond standard fixed-geometry assumptions (which artificially assume no thermal limit). Instead, it actively couples:

- **Core Gas Aerothermodynamics:** Solving 1D isentropic flow and the Bartz correlation to find the baseline convective heat transfer.
- **Liquid Film Cooling (Stechman & Hatch-Papell):** Calculating the vaporization and turbulent mixing of a liquid coolant layer to exponentially reduce the driving convective temperature.
- **Moving Boundary Heat Conduction (Stefan Problem):** Using a Landau Coordinate Transformation to solve the transient 1D radial heat equation in a shrinking domain as the liner ablates.

2. Core Gas Aerothermodynamics

Before evaluating boundary protection, the baseline state of the core combustion gas must be defined.

2.1. Isentropic Chamber Flow

The subsonic Mach number within the chamber (M_{ch}) is derived using the 1D choked flow Area-Mach relation [7]. For a known throat area (A_t) and chamber area (A_{ch}), M_{ch} is the root of:

$$\frac{A_t}{A_{ch}} = M_{ch}\left(\frac{2+(\gamma-1)M_{ch}^2}{\gamma+1}\right)^{-\frac{\gamma+1}{2(\gamma-1)}} \tag{2.1}$$

The free-stream static temperature (T_∞) and the adiabatic recovery temperature (T_r), which accounts for the kinetic energy recovery in a turbulent boundary layer using a recovery factor $r = Pr^{1/3}$, are calculated as:

$$T_\infty = \frac{T_{ch}}{1+\frac{\gamma-1}{2}M_{ch}^2} \tag{2.2}$$

$$T_r = T_\infty\left(1+ Pr^{\frac{1}{3}}\frac{\gamma-1}{2}M_{ch}^2\right) \tag{2.3}$$

where T_{ch} is the stagnation temperature of the combustion chamber.

2.2. Fluid Properties and Bartz Correlation

To correct for steep temperature gradients across the boundary layer, fluid properties are evaluated at the Eckert arithmetic mean temperature. Density and viscosity at the wall and free stream are:

$$\rho_\infty = \frac{P_{ch}}{R_g T_\infty} \tag{2.4}$$

$$\rho_{wall} = \frac{P_{ch}}{R_g T_r} \tag{2.5}$$

$$\rho_{am} = \frac{\rho_{wall} + \rho_\infty}{2} \tag{2.6}$$

Viscosity variation is modeled using a power-law extrapolation from reference throat conditions:

$$\mu_\infty = \mu_{ref} \left(\frac{T_\infty}{T_{ref}}\right)^\omega \tag{2.7}$$

$$\mu_{wall} = \mu_{ref} \left(\frac{T_r}{T_{ref}}\right)^\omega \tag{2.8}$$

$$\mu_{am} = \frac{\mu_{wall} + \mu_\infty}{2} \tag{2.9}$$

The core convective heat transfer coefficient (h_{ch}) is computed via the modified Bartz correlation [1]:

$$Re = \frac{\rho_\infty v_\infty D_{ch}}{\mu_\infty} \tag{2.10}$$

$$Nu = 0.026 Re^{0.8} Pr^{0.4} \left(\frac{\rho_{am}}{\rho_\infty}\right)\left(\frac{\mu_{am}}{\mu_0}\right) \tag{2.11}$$

$$h_{ch} = \frac{Nu\, k_{cond}}{D_{ch}} \tag{2.12}$$

3. Liquid Film Cooling Model

To protect the inner wall, a liquid mass flow fraction of fuel (e.g., RP-1) is injected along the perimeter. This model utilizes the Stechman aerodynamic correlation [2] combined with the Hatch-Papell thermal decay [3].

3.1. Injection Kinematics

The total mass flow rate is established by the choked flow equation at the throat:

$$\dot{m} = \frac{P_{ch} A_t}{\sqrt{R_g T_{ch}}} \sqrt{\gamma \left(\frac{2}{\gamma + 1}\right)^{\frac{\gamma+1}{\gamma-1}}} \tag{3.1}$$

The film mass flow rate is a user-defined fraction:

$$\dot{m}_{film} = \%_{film} \dot{m}_{tot} \tag{3.2}$$

Given N injection orifices of diameter $D_{injector}$, the total injection area is:

$$A_{injectors} = N\pi \left(\frac{D_{injector}}{2}\right)^2 \tag{3.3}$$

The initial liquid injection velocity (u_{init}) is:

$$u_{init} = \frac{\dot{m}_{film}}{\rho_{init} A_{injectors}} \tag{3.4}$$

Note: ρ_{init} is the liquid density evaluated at injection temperature T_{init} and chamber pressure P_{ch}.

3.2. Pseudo-Critical Thermodynamic State

As the liquid film absorbs heat at supercritical pressures, it does not boil but undergoes a sharp density and specific heat change. The model performs (by using CoolProp [5]) a thermal sweep to locate the pseudo-critical temperature T_{pc}, defined as the temperature where the specific heat (c_p) maximizes.

Once T_{pc} is found, the effective specific heat representing the energy required to bring the liquid from injection to the pseudo-critical vapor-like state is evaluated via the enthalpy jump:

$$c_{p,real} = \frac{H(T_{pc}, P_{ch}) - H(T_{init}, P_{ch})}{T_{pc} - T_{init}} \tag{3.5}$$

3.3. Stechman Aerodynamic Mixing

The physical thickness of the injected film if it were a uniform continuous slot is the equivalent slot thickness (S_{film}):

$$S_{film} = \frac{\dot{m}_{film}}{\rho_{film} u_{init} \pi D_{ch}} \tag{3.6}$$

Note: ρ_{film} is evaluated at T_{pc}.

The velocity ratio between the core free-stream gas and the injected film dictates the shear destruction of the boundary layer:

$$\beta = \frac{v_\infty}{u_{init}} \tag{3.7}$$

An empirical shape factor (f_β) accounts for the aerodynamic penalty of mismatched velocities:

$$f_\beta = \left(\frac{1}{\beta}\right)^{1.5\left(\frac{1}{\beta}-1\right)} \quad \text{if} \quad \beta \leq 1 \tag{3.8}$$

$$f_\beta = 1 + 0.4 \arctan(\beta - 1) \quad \text{if} \quad \beta > 1 \tag{3.9}$$

The final Stechman turbulent mixing parameter (ϕ) is:

$$\phi = \frac{1}{\left(S_{film}\beta\right)^{1/8} f_\beta} \tag{3.10}$$

3.4. Hatch-Papell Effective Temperature Decay

The interaction between the core gas and the film exponentially reduces the driving temperature for convection along the swept surface area.

$$A_{film} = \pi D_{ch} L_{ch} \tag{3.11}$$

The effective adiabatic wall temperature (T_{eff}) is:

$$T_{eff} = T_r - \left(T_r - T_{pc}\right) \exp\left(-\frac{h_{ch} A_{film}}{c_{p,real} \dot{m}_{film} \phi}\right) \tag{3.12}$$

Physics Note: In the subsequent solid thermal solver, convection is driven by ($T_{eff} - T_{wall}$), while radiation is completely unaffected by the transparent film and is driven by ($T_{ch}^4 - T_{wall}^4$).

4. Transient Heat Conduction & The Landau Transformation

4.1. The Moving Boundary Problem

The 1D radial unsteady heat conduction equation for the ablative liner (T_1) and the metallic casing (T_2) must be solved simultaneously. The metallic casing domain remains geometrically fixed and is governed by the standard radial heat equation in physical coordinates:

$$\frac{\partial T_2(r,t)}{\partial t} = \alpha_2 \frac{\partial^2 T_2(r,t)}{\partial r^2} + \frac{\alpha_2}{r}\frac{\partial T_2(r,t)}{\partial r} \tag{4.1}$$

where $\alpha_2 = \frac{k_2}{\rho_2 c_2}$ is the metallic casing thermal diffusivity.

However, because the liner ablates, its inner radius $s(t)$ increases, and its thickness $e_1(t)$ decreases over time. A Landau transformation maps this variable physical domain $r(t)$ to a fixed computational grid $\xi \in [0,1]$ [6]:

$$\xi = \frac{r - s(t)}{e_1(t)} \tag{4.2}$$

$$r_1(\xi, t) = s(t) + \xi e_1(t) \tag{4.3}$$

Where $\xi = 0$ is the receding inner surface, and $\xi = 1$ is the fixed interface with the metal casing.

Applying the chain rule, the spatial derivatives become:

$$\frac{\partial}{\partial r} = \frac{1}{e_1(t)}\frac{\partial}{\partial \xi} \tag{4.4}$$

$$\frac{\partial^2}{\partial r^2} = \frac{1}{[e_1(t)]^2}\frac{\partial^2}{\partial \xi^2} \tag{4.5}$$

The temporal derivative transforms to include a fictitious advection term driven by the recession velocity $\dot{s}(t)$:

$$\left.\frac{\partial T_1}{\partial t}\right|_r = \left.\frac{\partial T_1}{\partial t}\right|_\xi + \frac{\partial T_1}{\partial \xi}\frac{\partial \xi}{\partial t} \tag{4.6}$$

Since

$$\frac{\partial \xi}{\partial t} = \frac{-\dot{s}(t)e_1(t) - [r - s(t)]\dot{e_1}(t)}{[e_1(t)]^2} \tag{4.7}$$

and geometric constraints dictate that

$$\dot{e_1}(t) = -\dot{s}(t) \tag{4.8}$$

, this partial derivative reduces to

$$\frac{\partial \xi}{\partial t} = \frac{\dot{s}(t)(\xi - 1)}{e_1(t)} \tag{4.9}$$

Substituting these relationships into the radial heat equation yields the complete PDE for the liner in the fixed ξ computational domain:

$$\frac{\partial T_1(\xi,t)}{\partial t} = \frac{\alpha_1}{[e_1(t)]^2}\frac{\partial^2 T_1(\xi,t)}{\partial \xi^2} + \left[\frac{\dot{s}(t)(1-\xi)}{e_1(t)} + \frac{\alpha_1}{e_1(t)[s(t)+\xi e_1(t)]}\right]\frac{\partial T_1}{\partial \xi} \tag{4.10}$$

where $\alpha_1 = \frac{k_1}{\rho_1 c_1}$ is the liner thermal diffusivity.

Global Perspective: This mathematical formulation resolves the Stefan problem without the extreme computational cost of dynamically remeshing the spatial grid at every time step. By projecting the moving boundary into a convective transport term, specifically $\frac{\dot{s}(t)(1-\xi)}{e_1(t)}$, the PDE effectively advects the temperature profile inward precisely at the rate the material is consumed. The casing (T_2), not being subject to mass loss, is solved natively in physical space, while the two domains seamlessly exchange energy across a strictly enforced, non-moving numerical boundary at $\xi = 1\ (r = R_{J1})$.

4.2. Numerical Discretization (Upwind Scheme)

The transformed PDE for T_1 is solved using an Explicit Euler finite difference scheme. To prevent numerical oscillations from the advective term, an Upwind scheme is required.

Defining the diffusion and advection coefficients for a node j:

$$C_{diff} = \frac{\alpha_1}{[e_1(t)]^2} \tag{4.11}$$

$$C_{adv,j} = \frac{\dot{s}(t)(1-\xi_j)}{e_1(t)} + \frac{\alpha_1}{e_1(t) r_j} \tag{4.12}$$

The discrete time-marching equation dynamically shifts the spatial stencil based on the sign of $C_{adv,j}$:

- If $C_{adv,j} \geq 0$:

$$T_j^{n+1} = (1 - 2S_{diff} + S_{adv})T_j^n + (S_{diff} - S_{adv})T_{j-1}^n + S_{diff}T_{j+1}^n \tag{4.13}$$

- If $C_{adv,j} < 0$:

$$T_j^{n+1} = (1 - 2S_{diff} - S_{adv})T_j^n + S_{diff}T_{j-1}^n + (S_{diff} + S_{adv})T_{j+1}^n \tag{4.14}$$

where

$$S_{diff} = C_{diff}\frac{\Delta t}{\Delta \xi^2} \tag{4.15}$$

$$S_{adv} = C_{adv,j}\frac{\Delta t}{\Delta \xi} \tag{4.16}$$

4.3. Boundary and Interface Conditions

To conserve energy seamlessly without generating artificial heat, the interface linking the liner to the metal casing enforces continuity of both temperature and heat flux:

$$T_1(1,t) = T_2\big(r = R_{J1}, t\big) \tag{4.17}$$

$$\frac{k_1}{e_1(t)} \frac{\partial T_1(\xi,t)}{\partial \xi}\bigg|_{\xi=1^-} = k_2 \frac{\partial T_2(r,t)}{\partial r}\bigg|_{r=R_{J1}^+} \tag{4.18}$$

Numerically, this is explicitly solved using a conductance-weighted average:

$$T_{interface}^{n+1} = \frac{G_1 T_{liner,J1-1}^n + G_2 T_{casing,2}^n}{G_1 + G_2} \tag{4.19}$$

where

$$G_1 = \frac{k_1}{e_1 \Delta\xi} \tag{4.20}$$

$$G_2 = \frac{k_2}{\Delta r_2} \tag{4.21}$$

At the outer boundary of the metallic casing ($r = R_{J1} + e_2$), heat is lost to the ambient environment via natural convection:

$$-k_2 \frac{\partial T_2}{\partial r}\bigg|_{r=R_{J1}+e_2} = h_{ext}\big[T_2\big(R_{J1} + e_2, t\big) - T_\infty\big] + \sigma\epsilon_{ext}\big[T_2^4\big(R_{j1} + e_2, t\big) - T_{amb}^4\big] \tag{4.22}$$

5. Erosive Ablation & The Stefan Condition

At the exposed inner wall ($\xi = 0$), the incident heat flux combines convection and radiation. To couple this with the film cooling model, the convective driver is decoupled to use the effective adiabatic wall temperature (T_{aw}), while radiation relies on the core gas temperature (T_{rad}):

$$-\frac{k_1}{e_1(t)} \frac{\partial T_1(\xi,t)}{\partial \xi}\bigg|_{\xi=0} = h_{ch}(t)[T_{aw}(t) - T_1(0,t)] + \sigma\epsilon[T_{rad}^4(t) - T_1^4(0,t)] \tag{5.1}$$

The ablation regime is governed by the surface temperature relative to the material's ablation threshold (T_{abl}):

- **Non-Ablative Heating ($\boldsymbol{T_1(0,t) < T_{abl}}$):**

  The boundary operates under Neumann condition. No mass is lost ($\dot{s}(t) = 0$), and all incident heat conducts into the solid.

- **Active Ablation ($\boldsymbol{T_1(0,t) \geq T_{abl}}$):**

  The surface is mathematically clamped to the ablation temperature (Dirichlet condition). The incoming heat flux is partitioned between conducting into the solid and breaking chemical bonds to ablate mass.

The conductive flux penetrating the solid is defined strictly by the temperature gradient at the surface:

$$q_{cond}(t) := -\frac{k_1}{e_1(t)} \frac{\partial T_1(\xi, t)}{\partial \xi}\bigg|_{\xi=0} \tag{5.2}$$

The recession velocity $\dot{s}(t)$ is explicitly resolved using the Stefan condition, requiring the Effective Heat of Ablation (Q^*):

$$\dot{s}(t) = \frac{\max\{0,\ h_{ch}(t)[T_{aw}(t) - T_1(0,t)] + \sigma\epsilon[T_{rad}^4(t) - T_1^4(0,t)]\ - q_{cond}(t)\}}{\rho_1 Q^*} \tag{5.3}$$

The internal geometry is subsequently updated for the next step using the forward Euler approximation:

$$s^{n+1} = s^n + \dot{s}^{n+1}\Delta t \tag{5.4}$$

6. Stability Constraints

Because this model utilizes an explicit temporal integration scheme, the time step Δt must be strictly controlled to prevent divergence. The dynamic time step is restricted by both the Fourier (diffusion) and CFL (advection) numbers at the moving boundary:

$$\Delta t_{liner} \leq \left(\frac{2C_{diff}}{\Delta\xi^2} + \frac{C_{adv,\xi=0}}{\Delta\xi}\right)^{-1} \tag{6.1}$$

$$\Delta t_{casing} \leq \frac{\Delta r_2^2}{2\alpha_2} \tag{6.2}$$

The solver enforces

$$\Delta t = \beta \min\left(\Delta t_{liner},\ \Delta t_{casing}\right) \tag{6.3}$$

where β is a safety factor applied dynamically to ensure absolute numerical stability as the material ablates and $e_1(t)$ approaches zero.

7. Model Limitations and Assumptions

While this model provides a highly capable and numerically robust framework for transient heat transfer and ablation, it relies on several phenomenological simplifications that must be acknowledged when interpreting the results:

- **1D Radial Conduction Assumption:** The model strictly computes heat transfer in the radial direction. It neglects axial heat conduction, which can be significant near the injector plate or the nozzle throat, potentially leading to slightly conservative (hotter) temperature predictions in those specific regions.
- **Constant Material Properties:** Solid properties such as thermal conductivity and specific heat are treated as constants. In reality, ablative polymers and metal casings experience significant property variations as temperatures rise, particularly during pyrolysis and char formation.

- **Empirical Film Cooling:** The film cooling model assumes the liquid homogenizes and decays exponentially along the wall. It does not resolve complex 2D multi-phase fluid dynamics, such as droplet entrainment, film boiling, or boundary layer separation. These effects are instead lumped into the empirical Stechman shape factor.
- **Simplified Energetic Ablation:** The ablation logic relies entirely on a single macroscopic parameter, the Effective Heat of Ablation. It does not explicitly model the chemical kinetics of pyrolysis, the outgassing of pyrolysis gases (which would provide additional boundary layer blocking), or mechanical spallation.
- **Perfect Thermal Interface:** The conductance-weighted boundary assumes perfect thermal contact between the liner and the casing, ignoring any potential thermal contact resistance or gap formation due to differential thermal expansion.

8. Engineering Applications and Design Trade-Offs

The modular nature of this code, coupling core gas dynamics, film cooling, and a moving ablation boundary, makes it a highly versatile tool for iterative engine design. It is particularly valuable in dynamic laboratory environments where both liquid and solid propulsion systems are developed and tested.

8.1. Liquid Engine Film Cooling Optimization

For liquid rocket engines relying on regenerative or film cooling, injecting liquid coolant creates a massive thermal buffer but introduces a performance penalty. The coolant does not combust at optimal mixture ratios, reducing the overall Specific Impulse (Isp). This model allows engineers to perform rapid parametric sweeps to find the optimal Coolant Mass Fraction. By iteratively adjusting the film percentage, the designer can identify the exact minimum coolant flow required to keep the wall temperature below the material's yield or melting point, thereby maximizing Isp without sacrificing structural integrity.

8.2. Solid Rocket Motor Liner Sizing

For solid rocket motors, the casing must be protected from the hot combustion gases until the burn out. Oversizing the ablative liner adds parasitic dead mass, reducing the mass fraction and ΔV of the vehicle. Under sizing it leads to catastrophic casing failure.

By running this model over the expected burn time, engineers can track the recession velocity and the final remaining liner thickness. This enables the precise calculation of the minimum required initial liner thickness plus a defined safety margin, minimizing inert mass while guaranteeing the casing never exceeds its thermal limits.

9. Final Conclusions

The integration of a Landau coordinate transformation for the moving boundary problem, coupled with aerothermodynamic boundary conditions (Bartz and Stechman/Hatch-Papell), yields a highly efficient and rigorous 1D thermal solver.

By explicitly calculating the erosion of the liner and the exponential thermal decay of liquid film cooling, this model bridges the gap between purely empirical steady-state estimates and computationally expensive Computational Fluid Dynamics (CFD) or Finite Element Analysis (FEA). It provides aerospace engineers with a fast-solving, physics-based tool capable of evaluating transient thermal loads, sizing protective liners, and optimizing film cooling mass fractions with a high degree of confidence.

10. Phenomenological Analysis of System Dynamics and Thermal Behavior

This section provides an analytical evaluation of the system dynamics and transient thermal behavior based on the computational results. The simulated outputs serve as a baseline stress test to evaluate the survivability of an uncooled ablative wall, providing the quantitative rationale for subsequent design iterations.

10.1. Aerothermodynamic Trade-Off Evaluation (Film Cooling)

The initial analytical module functions as a preliminary sizing tool. It evaluates the necessity of implementing active thermal mitigation prior to executing the transient numerical solver.

The following input parameters are used to evaluate the viability of the liquid film cooling:

```
%% ----------------------------------------------------------------------
%% 1) SIMPLIFIED INPUTS (Main Combustion Gas & Geometry)
%% ----------------------------------------------------------------------
% Simplified chamber geometry
D_ch = 53;       % Chamber / convergent inlet diameter [mm]
D_t  = 26.5;       % Throat diameter [mm]
% Conversion to meters
D_ch = D_ch * 1e-3;
D_t  = D_t  * 1e-3;
% Combustion Gas Thermodynamic Properties (From NASA CEA)
Rg = 436.545;       % Specific gas constant at throat [J/(  [4]  ]
Pr = 0.5590;        % Prandtl number
mu_0_mP = 0.87427;  % Stagnation chamber viscosity [mP]
w = 0.7125;         % Viscosity power-law adjustment parameter
mu_ref_mP = 0.81359;% Reference viscosity at throat [mP]
T_ref = 2603.62;    % Reference static temperature at throat [K]
k_cond_ref = 0.31615; % Reference thermal conductivity at throat [W/(m*K)]
Tch = 2877.68;      % Stagnation chamber temperature [K]
Pch = 26.641 * 1e5; % Stagnation chamber pressure [Pa]
k_ch = 1.2104;      % Chamber isentropic coefficient (gamma)
```

```
%% ---------------------------------------------------------------------
%% 6) FILM COOLING (STECHMAN & COOLPROP INTEGRATION)
%% ---------------------------------------------------------------------
% Film Cooling geometric and injection inputs [ADJUST THESE VALUES]
L_ch = 195.58e-3;        % Cylindrical chamber length [m]
N_orificios = 40;        % Number of injection orifices
D_orificio = 1e-3;       % Injection orifice diameter [m]
porcentaje_film = 0.05; % Film cooling mass fraction (e.g., 0.05 = 5%)
T_init = 293.15;         % RP-1 injection temperature [K]
fluid = 'n-Dodecane';    % CoolProp surrogate fluid for RP-1
```

Figure 10.1: MATLAB input parameters for the Stechman film cooling and Bartz convective models.

Upon execution of the Stechman aerodynamic model coupled with thermodynamic properties, the following operational results are obtained:

```
=====================================================================
  A) MAIN GAS IN CHAMBER (BARTZ MODEL)
=====================================================================
 Mach Number (M_ch):              0.14967
 Core Velocity (v_inf):           184.34 m/s
 Static Temp (T_inf):             2870.91 K
 Recovery Temp (Tr):              2876.49 K
 Convective Coeff (h_ch):         2602.34 W/(m^2*K)
---------------------------------------------------------------------
 >>> IF NO FILM COOLING: USE T_aw0 = 2876.49 K | T_rad0 = 2870.91 K | h0 = 2602.34 W/(m^2*K) <<<
=====================================================================

=====================================================================
  B) FILM COOLING (STECHMAN & COOLPROP)
=====================================================================
 Injected Mass Flow:              0.0426 kg/s
 Injection Velocity (u_init):     1.81 m/s
 Velocity Ratio (beta):           102.03
 Pseudo-critical Temp (Tpc):      688.94 K
 Effective Specific Heat (Cp):    3034.45 J/(kg*K)
 Mixing Parameter (phi):          0.8770
---------------------------------------------------------------------
 >>> IF FILM COOLING:    USE T_aw0 = 1839.85 K | T_rad0 = 2870.91 K | h0 = 2602.34 W/(m^2*K) <<<
=====================================================================
```

Figure 10.2: Numerical console output comparing bare-wall convective drivers and mitigated film-cooled boundary conditions.

- **Baseline vs. Mitigated Convective Drivers:** The numerical output quantifies the severity of the aerothermodynamic environment. For the baseline uncooled configuration, the boundary layer recovery temperature imposes a convective driving temperature of $T_r = 2876.49\,K$ Conversely, introducing a 5% liquid coolant mass fraction mitigates this condition. Despite the shear degradation of the boundary layer, parameterized by the Stechman mixing coefficient ($\phi = 0.8770$), the effective adiabatic wall temperature (T_{aw0}) is drastically reduced to $1839.85\,K$.
- **Design Implications:** This establishes a clear performance trade-off. Implementing film cooling imposes a 5% penalty on the Specific Impulse but reduces the convective thermal potential by approximately 61%. To justify this efficiency penalty, the baseline configuration (uncooled wall) must first be evaluated to determine whether the structural thermal limits are exceeded.

10.2. Baseline Transient Analysis: Unmitigated Ablation

The explicit Euler solver evaluates the baseline configuration: a 2 mm ablative liner subjected to the unmitigated convective driving temperature ($T_{aw0} = 2870.91\ K$). For this 12 second transient study, the following material properties and boundary conditions have been introduced:

```
%% ---------------------------------------------------------------------
%% 1) USER-ADJUSTABLE PARAMETERS (material properties, BCs, geometry)
%% ---------------------------------------------------------------------
% --- liner (inner ablative layer) ---
rho1   = 1720;       % kg/m^3
c1     = 1250;       % J/(kg*K)
k1     = 0.6;        % W/(m*K)
alpha1 = k1 / (rho1 * c1);

% --- casing (outer metallic layer) ---
rho2   = 7850;       % kg/m^3
c2     = 480;        % J/(kg*K)
k2     = 43;         % W/(m*K)
alpha2 = k2 / (rho2 * c2);

% Radiation
sigma       = 5.670374419e-8;  % Stefan-Boltzmann [W/(m^2*K^4)]
epsilon     = 0.9;             % emissivity (inner liner)
epsilon_ext = 0.85;            % emissivity (outer metallic casing)

% Temperatures and ablation parameters
T_amb  = 293;        % ambient temperature [K]
T_abl  = 1950;       % ablation threshold temperature [K]
Q_star = 10e6;       % effective heat of ablation [J/kg] (tune as needed)

% Convective coefficients and chamber temperatures (From Film Cooling Model)
h_ext  = 30;         % external convection [W/(m^2*K)]
h0     = 2602.34;    % inner convection coeff (h_ch from Bartz) [W/(m^2*K)]
T_aw0  = 2870.91;    % Film Cooling effective temp (for Convection) [K]
T_rad0 = 2870.91;     % Core gas static temp (T_inf) for accurate Radiation [K]
%% ---------------------------------------------------------------------
%% 2) GEOMETRY & SPATIAL DISCRETIZATION
%% ---------------------------------------------------------------------
D_int   = 53e-3;         % inner diameter [m]
e1_init = 2e-3;          % initial liner thickness [m]
R_int0  = D_int / 2;     % initial inner radius [m]
R_J1    = R_int0 + e1_init;  % global location of liner/casing interface [m]
e2      = 3.6e-3;        % casing thickness [m]

% Nodes
J1 = 50;                % nodes in liner (xi-space)
Jc = 100;               % nodes in casing (r-space)
```

Figure 10.3: Input parameters and boundary conditions for the transient 1D radial heat conduction and ablation solver.

The subsequent subsections detail the transient response of this system throughout the integration period.

10.2.1. Non-Linear Ablation Kinetics

A standard assumption in simplified thermal models is a constant ablation rate. However, the numerical resolution of the Stefan problem demonstrates highly non-linear recession behavior.

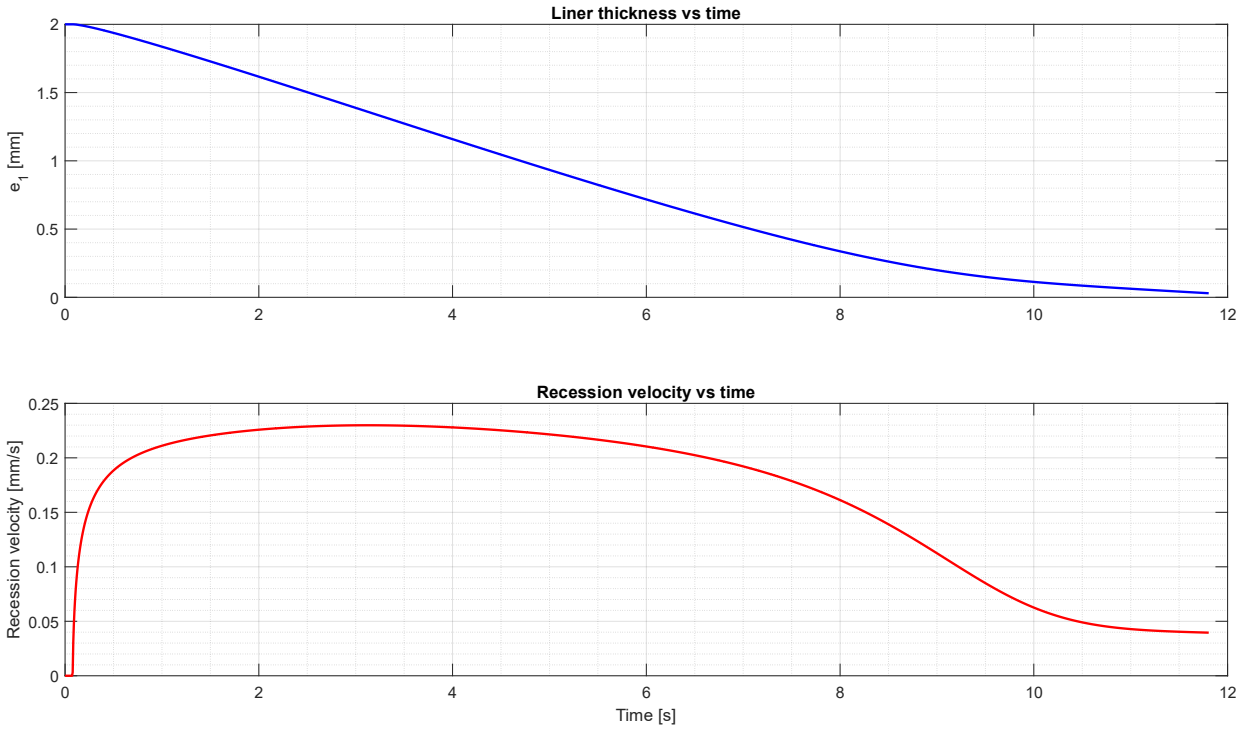

Figure 10.4: Transient evolution of the ablative liner thickness and its corresponding non-linear recession velocity.

- **Ignition Transient:** Upon motor ignition, the inner surface rapidly reaches the phase-change threshold ($T_{abl} = 1950\ K$). Because the initial thermal resistance of the cold liner is high, the internal conductive flux (q_{cond}) remains negligible. Consequently, the net incident heat flux is almost entirely expended on the heat of ablation (Q^*), resulting in a maximum initial peak in the recession velocity.
- **Equilibrium Shift and Self-Limiting Behavior:** As time progresses, the recession velocity exhibits a distinct exponential decay. This highlights the self-limiting nature of the ablative process. As the liner thickness $e_1(t)$ decreases, the proximity of the highly conductive metallic casing steepens the thermal gradient across the remaining ablative material. This geometric change increases the inward conductive flux (q_{cond}), thereby reducing the net energy available for polymer pyrolysis and mass removal. A thinner liner ablates at a lower rate, but exponentially increases the thermal load transferred to the structural casing.

10.2.2. Numerical Stiffness and Interface Energy Conservation

Analyzing the flux continuity at the material interface validates the robustness of the Landau transformation applied to the moving boundary problem.

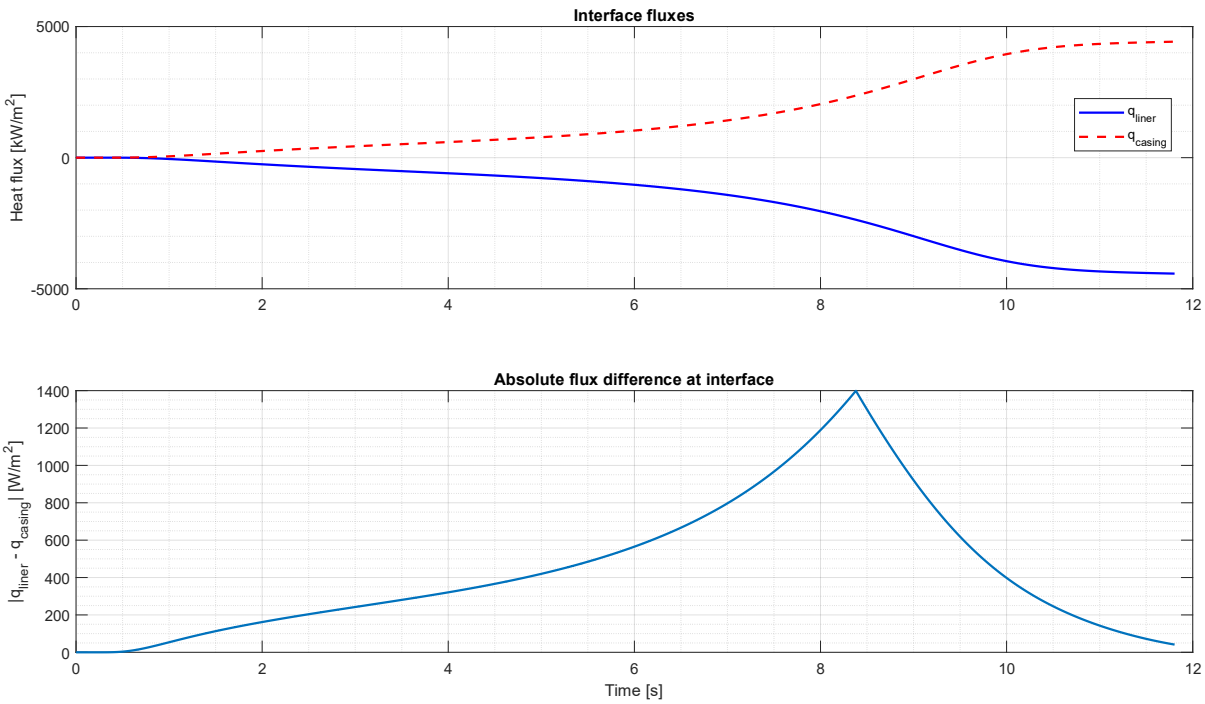

Figure 10.5: Energy conservation analysis showing heat flux continuity at the moving boundary interface and the associated numerical error.

- **Asymptotic Error Manifestation:** The absolute flux difference graph indicates a localized peak in numerical error toward the end of the simulation. This is a manifestation of mathematical stiffness. As the liner thickness approaches zero ($e_1(t) \to 0$), the diffusive (α_1/e_1^2) and advective coefficients in the transformed Partial Differential Equation (PDE) approach singularity.
- **Validation of the Upwind Scheme:** Despite this extreme mathematical tension at the moving boundary, the explicit Upwind finite difference scheme maintains energy conservation within acceptable relative tolerances. The peak interface flux error remains negligible compared to the total incident heat flux, validating the solver's stability without incurring the severe computational cost of dynamic spatial remeshing.

10.2.3. Thermal Inertia and Heat Wave Propagation

The representative temperature evolution profiles demonstrate the temporal effects of the material's thermal diffusivity (α_1).

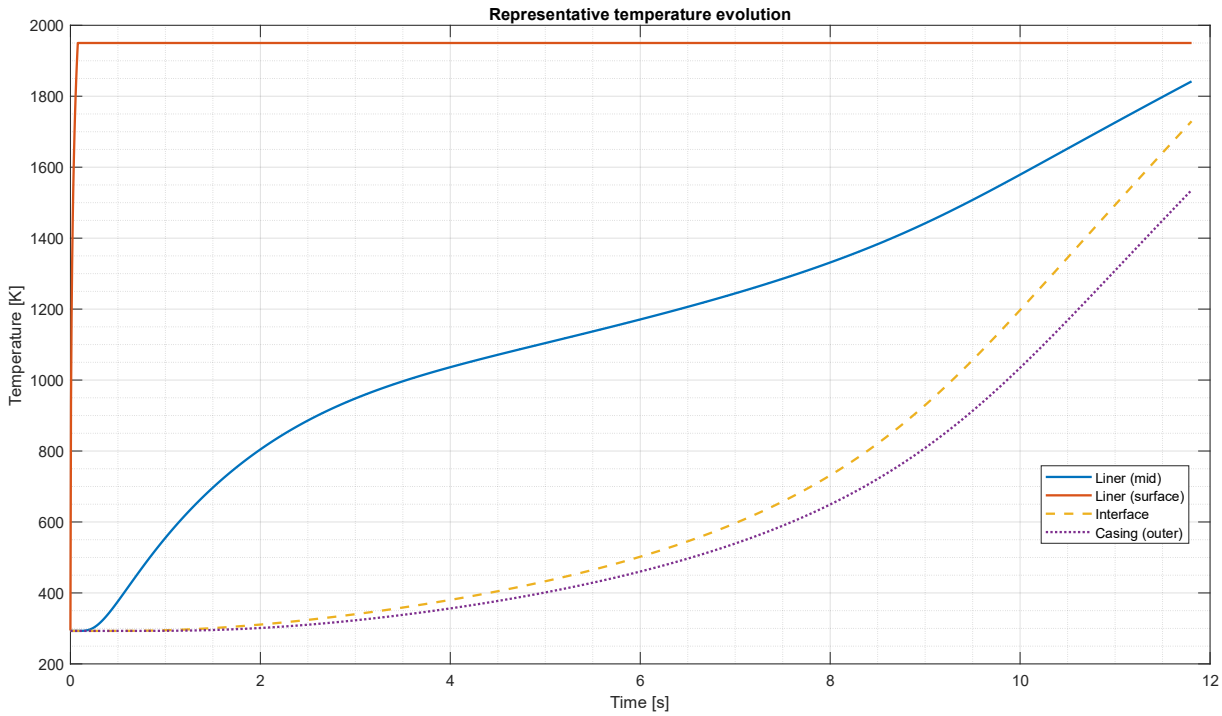

Figure 10.6: Transient temperature profiles demonstrating thermal inertia across the liner and subsequent heat wave propagation into the casing.

- **Thermal Lag:** A distinct temporal delay is observed between the inner surface reaching T_{abl} and the outer casing experiencing a temperature rise. During the initial seconds of the burn, the incident energy is entirely stored within the specific heat capacitance of the liner.
- **Structural Implications:** As the protective layer is consumed and its thermal resistance approaches zero, the heat wave propagates rapidly into the metallic casing. By $t \sim 12\ s$, the outer casing temperature exceeds $1400\ K$. For standard aerospace-grade alloys, this temperature far exceeds the annealing point and yield strength limits, indicating an inevitable structural failure under operational chamber pressures.

10.2.4. Final Thermal State and Design Iteration

The spatial temperature distribution at $t = 12\ s$ provides the definitive quantitative assessment of the baseline design.

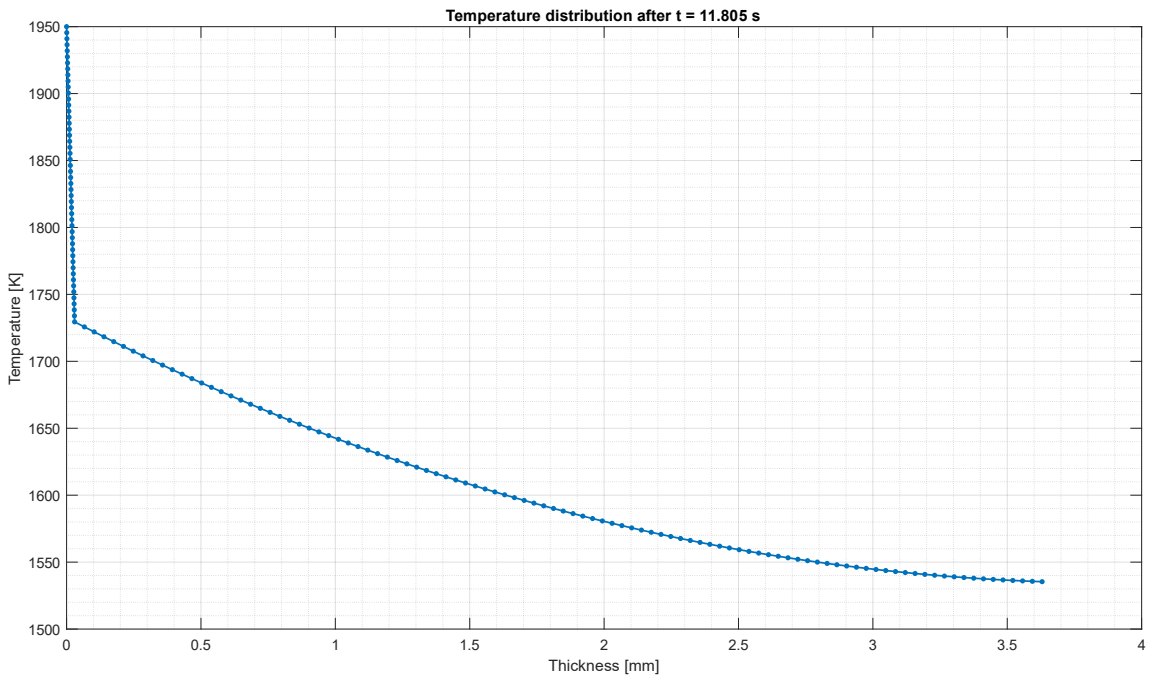

Figure 10.7: Final spatial temperature distribution across the residual liner and metallic casing at motor burnout (12 seconds).

- **Depleted Thermal Margins:** At the end of the integration period, the spatial profile shows a nearly depleted liner with a residual thickness of just 0.057 mm. The material interface rests at 1564 K, while the metallic casing operates almost isothermally near 1400 K due to its high thermal conductivity (low Biot number regime).
- **Engineering Conclusion:** The baseline configuration fails to satisfy structural safety factors. The analytical data dictates two primary design iterations:
  - **Increase Inert Mass:** Increase the initial liner thickness ($e_{1,init} \approx 3.5 - 4\ mm$) to maintain sufficient thermal resistance throughout the 12 seconds' burn.
  - **Implement Active Cooling:** Introduce the 5% liquid film cooling evaluated in Section 10.1. By suppressing the convective driving temperature to 1121 K, well below the 1950 K ablation threshold, the liner would cease to ablate entirely, functioning strictly as a solid insulator and preserving the structural integrity of the casing.

11. Fully Coupled Dynamic Aerothermodynamics

Previous sections evaluated the system under the assumption of a static aerothermodynamic environment, where the convective driving temperature (T_{aw}) and the heat transfer coefficient (h_{ch}) remained constant throughout the burn. However, physical rocket engines exhibit a highly non-linear feedback loop. To capture this, the numerical solver was upgraded into a fully coupled Digital Twin, resolving the interdependent physics at every time step.

11.1. The Non-Linear Feedback Loop

As the ablative liner erodes, the inner radius $s(t)$ recedes, actively increasing the internal chamber diameter $D_{ch}(t)$. Assuming the nozzle throat area (A_t) remains constant (e.g., using a heat-sink copper throat), the geometric expansion of the chamber dictates a continuous drop in the area ratio $A_t/A_{ch}(t)$.

This physical expansion leads to a sequence of coupled aerodynamic phenomena:

- **Flow Deceleration:** The subsonic Mach number $M_{ch}(t)$ decreases, analytically resolved at each time step using a Newton-Raphson root-finding algorithm over the 1D isentropic choked flow relations.
- **Thermal Relaxation:** A lower Mach number yields lower free-stream velocities (v_∞), reducing the kinetic energy available for recovery in the turbulent boundary layer. Consequently, the static temperature (T_∞) and recovery temperature (T_r) drop.
- **Convective Degradation:** The reduction in fluid velocity lowers the local Reynolds number, which directly reduces the Nusselt number and the Bartz convective heat transfer coefficient $h_{ch}(t)$.
- **Film Cooling Feedback:** For cooled configurations, the expanding perimeter increases the swept area (A_{film}) while the gas-to-liquid velocity ratio (β) drops, dynamically altering the Stechman mixing parameter (ϕ) and the effective film temperature T_{eff}.

11.2. Dynamic System Response

The real-time coupling of these equations yields a highly dynamic system response, mitigating the overly conservative estimates of uncoupled steady-state models.

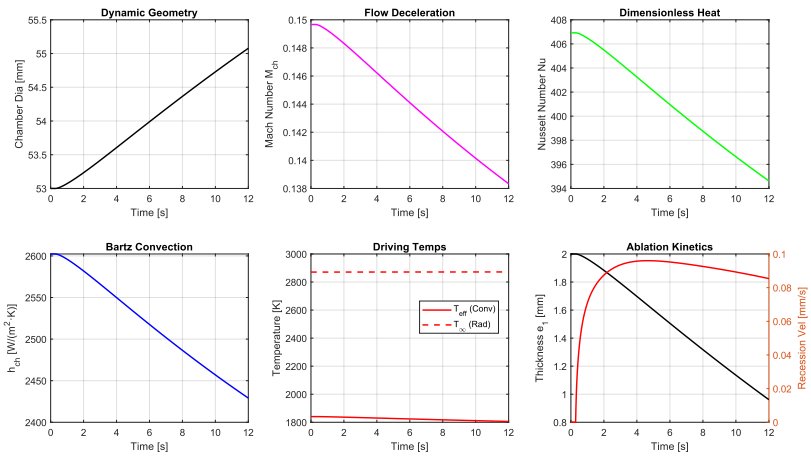

Figure 11.1: Transient evolution of the coupled aerothermodynamic variables during a 12-second burn with film cooling.

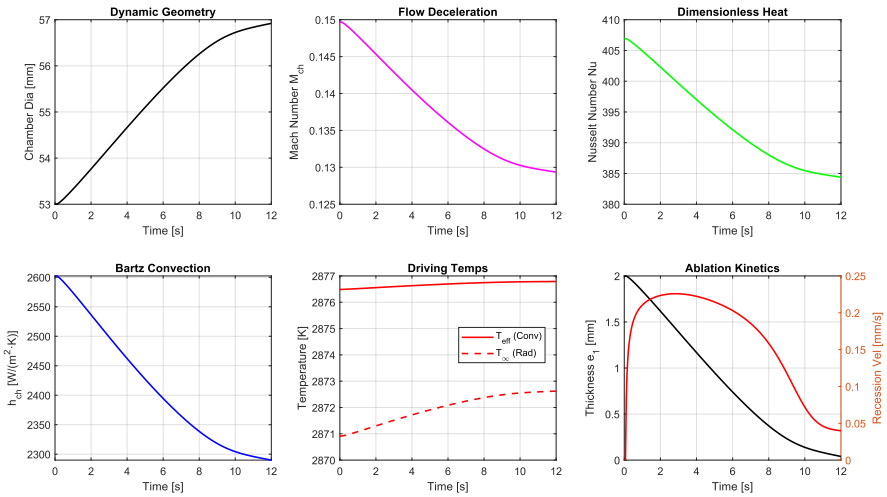

Figure 11.2: Transient evolution of the coupled aerothermodynamic variables during a 12-second burn without film cooling.

As observed in Figure 11.1, the ablation-driven geometry changes significantly alter the thermal environment:

- **Geometric Expansion:** Over the 12-second integration period, the chamber diameter expands continuously due to ablation.
- **Aerodynamic Relief:** The Mach number exhibits a steady deceleration, driving a proportional decline in the dimensionless Nusselt number.
- **Convective Drop:** The Bartz convective coefficient (h_{ch}) drops from an initial peak of approximately 2600 W/(m²·K) down to 2430 W/(m²·K). This confirms that as the motor burns and the chamber widens, the severity of the convective thermal load naturally relaxes, providing a slight self-preservation mechanism to the remaining liner.

11.3. Comparative Ablation Kinetics: The Engineering Trade-Off

To quantify the absolute value of the protective liquid layer, the solver's macro-structure evaluates and compares the ablation kinetics of a bare wall against a film-cooled wall under identical initial conditions.

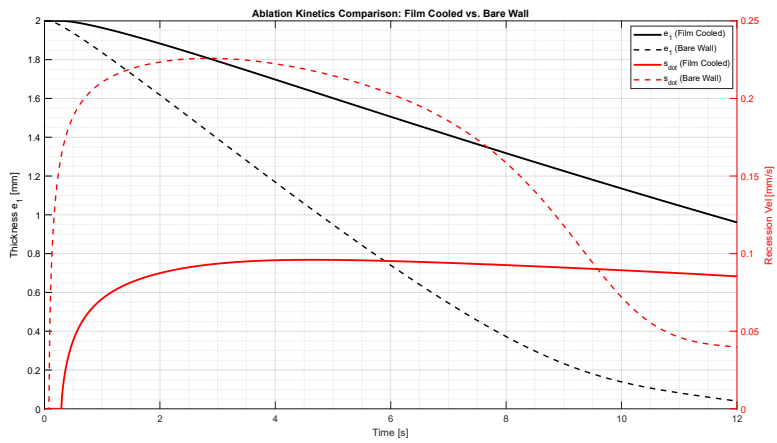

Figure 11.3: Comparative ablation kinetics between film-cooled (solid lines) and bare-wall (dashed lines) configurations.

Figure 11.3 highlights the critical engineering trade-offs regarding structural survivability:

- **Unmitigated Failure (Bare Wall):** The dashed lines represent the uncooled casing. Exposed to the full recovery temperature, the recession velocity peaks aggressively above 0.2 mm/s. By the end of the burn, the 2 mm liner is almost entirely consumed (approaching 0 mm thickness), ensuring critical thermal exposure and likely structural failure of the outer metal casing.
- **Mitigated Survivability (Film Cooled):** The solid lines demonstrate the effect of a 5% liquid coolant mass fraction. The recession velocity is heavily suppressed, plateauing near 0.1 mm/s. Consequently, the liner retains nearly 1 mm of virgin material at engine burnout.

Conclusion: The coupled model definitively proves that the aerodynamic penalty incurred by the Stechman mixing process is vastly outweighed by the thermal benefits. The active film cooling not only preserves 50% of the ablative liner but guarantees the thermal integrity of the structural casing.

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